

popquiz Application newton III

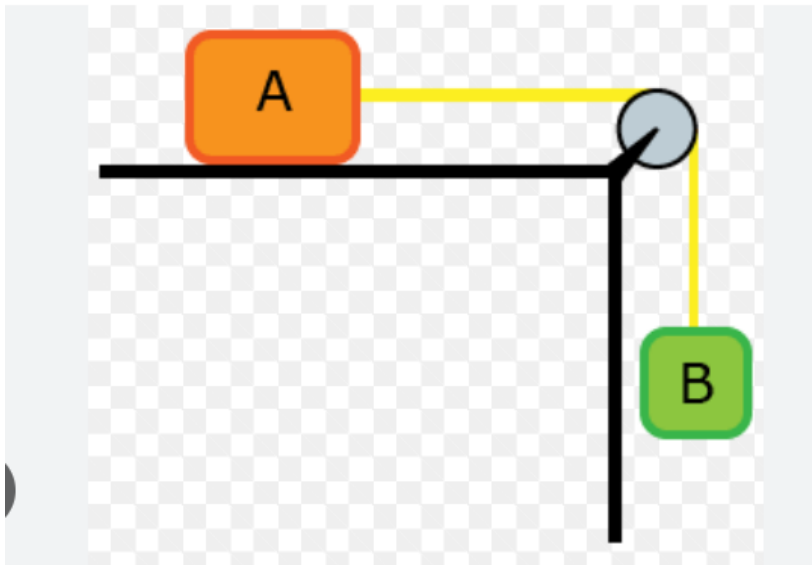
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Started: Apr 29 at 7:35pm

Quiz Instructions



Question 1 25 pts



A is on a surface with a kinetic coefficient of 0.2. A has a mass of 5kg. B has a mass of 10kg. What is the acceleration? take $g=10\text{m/s/s}$ **round to the nearest one**

hint:

mass A : friction left and tension right. tension is winning. kinetic friction = $\mu_k \times \text{normal}$ and normal=weight

mass B : weight down and tension up. weight is winning.

2 systems with 2 equations



7 m/s/s



3m/s/s



10m/s/s



6m/s/s



Question 2 20 pts

go to:

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html ↗ (https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html)

click on acceleration. check everything that can be checked. place the mouse on the dummy and have it push the crate. go slowly. static friction is opposing the applied force. What is the maximum static friction before the crate starts to move and kinetic friction takes over?

☐

about 255N

☐

about 55N

☐

about 125N

☐

about 105N



Question 3 20 pts

Building on the previous question.

static friction max = $\mu_s \times \text{normal force}$

μ_s is the coefficient of static friction.

what is μ_s here? its dimensionless.

along the vertical: up=down. so whats the normal ?

☐

0.25

☐

0.5

☐

0.1

☐

2.5



Question 4 25 pts

Now have the dummy move the crate. The kinetic friction is about

☐

about 20 N

☐

about 10N

☐

about 90N

☐

about 50N



Question 5 25 pts

Have 2 crates on top of each other. Get the system of move. Whats the kinetic frictional force?

☐

255N

☐

94N

☐

188N

☐

125N

Not saved

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